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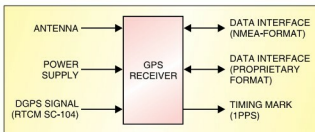


Fig. 6: GPS receiver block diagram

appropriate commands are given to the motor controllers.

Waypoint navigation. The GPS unit was upgraded to Garmin GPS 18 OEM. It provides non-volatile memory for storage of configuration information, a real-time clock and raw measurement output data in NMEA 0183 format (industry standard).

Each GPS data set is formed in the same way and has the following structure:

```
$GPRDTS, Inf_1, Inf_2, Inf_3, Inf_4,
```

```
Inf_5, Inf_6, Inf_n*CS<CR><LF>
```

For example, a GLL data set is shown

below:

```
$GPGLL, 4717.115,N, 00833.912,E, 130305
```

```
.0,A*32<CR><LF>
```

Data extraction. The UGV employs a Brute force algorithm which relies on the map to navigate. This algorithm evaluates the safest and shortest path to a given location based on cost calculations, which is determined from the frequency of encountering the obstacles. For navigation, the desired location is always a GPS waypoint.

After data extraction, the coordinates are obtained and the destination coordinates fed manually into a sub-controlling program. The program continuously calculates the relative angle between the destination point and the present direction of the vehicle. It decides which point is closest to its present location, selects that point, and then makes its angle calculations using Haversine formula:

$$a = \sin^2(\Delta lat/2) + \cos(lat_1) \cdot \cos(lat_2) \cdot \sin^2(\Delta long/2)$$

$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$d = R \cdot c$$

where 'R' is the earth's radius (mean radius = 6371 km), $\Delta lat = lat_2 - lat_1$, and $\Delta long = long_2 - long_1$

This data is returned continuously to the main controlling program where decisions are made. Once the point is reached, the next nearest uncovered point is selected. The process continues till the sixth waypoint is reached.

The GPS receiver returns coordinates, which are converted into decimal degrees. The data input from the GPS software and obstacle-detection sensor is interfaced in the main controlling program. The UGV then finds its way to the waypoints and continues to the next ones. The GPS data is updated continuously. ●

The author is pursuing BE in electronics and communication from M.S. Ramaiah Institute of Technology, Bengaluru